Mathematical Modeling

Sheet 1: Dual Simplex and Primal Recovery

Tasks 4 and 5(a)–5(d)

Personal study notes

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1 Task statements

4. Rozwiąż algorytmem dualnym. Na tej podstawie podaj rozwiązanie zagadnienie dualnego.

$$\begin{aligned} \min \quad & x_1 + 6x_2 + 2x_3 + 3x_4, \\ & 4x_1 + 2x_2 - x_3 + x_4 \geq 4, \\ & x_1 + x_2 + x_3 + x_4 \geq 5, \\ & -2x_1 - 2x_2 - x_3 - x_4 \geq -8, \\ & x_1, x_2, x_3, x_4 \geq 0. \end{aligned}$$

5.

- Podaj postać zadania dualnego ZD
- Rozwiąż ZD dwufazową metodą sympleks i podaj jego znalezione RO oraz optymalną wartość funkcji celu dla ZD
- Na podstawie znalezionego rozwiązania RO dla zadania ZD, oblicz RO dla zadania ZP (bez rozwiązywania ZP) oraz sprawdź optymalną wartość funkcji w ZP
- Czy zadanie ZD posiada rozwiązania alternatywne? Uzasadnij

(a)

$$\begin{aligned} \min \quad & 3x_1 + 10x_2 + 2x_3, \\ & -x_1 + 2x_2 + x_3 \geq 2, \\ & -x_1 + x_2 + x_3 \geq 1, \\ & -x_1 + 2x_2 - x_3 \geq -3, \\ & -x_1 + 2x_2 + x_3 \geq 1, \\ & x_1 \leq 0, \quad x_2 \geq 0, \quad x_3 \geq 0. \end{aligned}$$

(b)

$$\begin{aligned} \min \quad & 3x_1 + 10x_2 + x_3, \\ & -2x_2 - 2x_3 \geq -2, \\ & 2x_1 - x_2 \geq 1, \\ & -2x_1 + x_2 + x_3 \geq -5, \\ & 2x_1 + 2x_2 + x_3 \geq 1, \\ & x_1 \geq 0, \quad x_2 \in \mathbb{R}, \quad x_3 \geq 0. \end{aligned}$$

(c)

$$\begin{aligned} \min \quad & -x_1 + 5x_2 + 8x_3, \\ & -x_1 - x_2 + 2x_3 \geq -1, \\ & -x_1 + 3x_3 \geq 5, \\ & -x_1 + x_2 + x_3 \geq 1, \\ & -x_2 - 2x_3 \geq -1, \\ & x_1 \in \mathbb{R}, \quad x_2 \in \mathbb{R}, \quad x_3 \geq 0. \end{aligned}$$

(d)

$$\begin{aligned} \min \quad & x_1 - 5x_2 + 8x_3, \\ & 3x_1 + x_2 + 2x_3 \geq 1, \\ & -x_1 + x_3 \geq 5, \\ & -x_1 - x_2 + x_3 \geq 1, \\ & 3x_1 - x_2 - 2x_3 \geq -2, \\ & x_1 \geq 0, \quad x_2 \in \mathbb{R}, \quad x_3 \geq 0. \end{aligned}$$

2 Notation, domains, and the mathematical plan

All decision variables are continuous and obey the displayed sign restrictions. The data specify abstract linear programs, with no physical units supplied. In particular, a free variable may be negative: it must not acquire an extra nonnegativity restriction during conversion or verification.

The prerequisites are standard-form equivalence and bases in Chapter 02, Phase I and reduced costs in Chapter 03, and mixed-domain duality, complementarity, and dual simplex in Chapter 04. We apply these ideas to the complete tasks below.

For any equality-form maximization, use

$$\max c^T u, \qquad Mu = d, \quad u \geq 0.$$

A nonsingular basis B gives

$$\bar{M} = M_B^{-1} M, \quad \bar{d} = M_B^{-1} d, \quad \lambda = M_B^{-T} c_B, \quad q = M^T \lambda - c.$$

The tableau displays the equations $\bar{M}u = \bar{d}$ and the last row $q^T u = z_0 - c^T u$, where $z_0 = c_B^T \bar{d}$. Thus

$$c^T u = z_0 - \sum_{j \notin B} q_j u_j. \tag{1}$$

The last RHS is z_0, not a basic variable. Ratios are calculated in the column to the right of the RHS; $\longrightarrow$ in the separate Out column marks the chosen leaving row. The indicator row is labelled $z_j - c_j$; q_j abbreviates this difference in formulas. Vertical rules separate c_B, the basic-variable labels, the coefficient columns, and the RHS. Variable-column headings keep their order across pivots; only the basic-variable labels change. Artificial columns are removed after Phase I without reordering the retained columns. The c_B column gives the objective coefficient of each basic variable. The z_j row sums each constraint column with these weights; subtract the displayed c_j row to obtain $q_j = z_j - c_j$. The same weighted sum on the RHS gives z_0. In Phase I these are the auxiliary objective coefficients; in Phase II they are the original ones. Nonnegative indicators certify an upper bound on the maximum. The original minimum in task 4 is the negative of its transformed maximum. Tasks 5 maximize their dual objectives directly.

2.1 Why the domains determine the dual inequalities

Write a primal minimum as $\min c^T x$, $Ax \geq b$. A nonnegative multiplier y gives $y^T Ax \geq y^T b$. To make $c^T x \geq y^T Ax$ hold for every allowed x, column j must obey

Primal domain	Dual condition
$x_j \geq 0$	$a_j^T y \leq c_j$
$x_j \leq 0$	$a_j^T y \geq c_j$
$x_j \in \mathbb{R}$	$a_j^T y = c_j$

For example, multiplying a nonpositive x_j by $c_j - a_j^T y \leq 0$ gives a nonnegative product. A free variable can take either sign, so its coefficient must vanish. The resulting dual maximizes $b^T y$ over $y \geq 0$ with these column conditions. Negative entries in b do not change multiplier domains.

For a feasible pair, the gap has the decomposition

$$c^T x - b^T y = (c - A^T y)^T x + y^T (Ax - b) \geq 0. \tag{2}$$

Every term on the right is nonnegative with the appropriate domain convention. Optimal pairs make each term zero. This explains complementary slackness, rather than treating it as an additional system to memorize.

2.2 Two different simplex invariants

Primal simplex keeps $\bar{d} \geq 0$. It chooses an entering column with $q_k < 0$ and preserves feasibility by the ratio test

$$\theta = \min_{\bar{m}_{ik} > 0} \frac{\bar{d}_i}{\bar{m}_{ik}}.$$

Our Phase I and II paths choose the most negative indicator, breaking ties by smaller column index, then the smallest ratio, breaking ties by smaller leaving variable index. These choices specify these finite paths; they are not a general anti-cycling guarantee.

Dual simplex keeps $q \geq 0$. It first selects a negative RHS row r. Eligible entering columns have $\bar{m}_{rk} < 0$, and

$$\alpha = \min_{\bar{m}_{rj}<0} \frac{q_j}{-\bar{m}_{rj}}$$

preserves nonnegative indicators because $q'_j = q_j + \alpha \bar{m}_{rj}$. We select the most negative RHS and the smaller entering column in a tie. The update $z'_0 = z_0 + \alpha \bar{d}_r$ lowers the dual upper bound on a maximum. It does not describe improvement of an infeasible primal point.

Phase I maximizes minus the sum of nonnegative artificial variables. Its optimum is at most zero. A negative optimum certifies infeasibility; zero permits Phase II after artificials are removed and the original objective is recomputed. All successful paths here remove artificials from the basis, so no zero-basic artificial cleanup is needed.

3 Task 4: a complete dual-simplex solve

Put $s_i = (Ax - b)_i \geq 0$, negate each row, and maximize the negative cost:

$$\begin{aligned} \max \quad & -x_1 - 6x_2 - 2x_3 - 3x_4, \\ -4x_1 - 2x_2 + x_3 - x_4 + s_1 &= -4, \\ -x_1 - x_2 - x_3 - x_4 + s_2 &= -5, \\ 2x_1 + 2x_2 + x_3 + x_4 + s_3 &= 8. \end{aligned}$$

The surplus basis $B = (s_1, s_2, s_3)$ is nonsingular, with indicators $(1, 6, 2, 3, 0, 0, 0) \geq 0$. Its negative RHS entries make it primal infeasible, but exactly the dual-feasible start we need.

c_B	Basic	x_1	x_2	x_3	x_4	s_1	s_2	s_3	RHS	Dual ratios ($r = 2$)	Out
	c_j	-1	-6	-2	-3	0	0	0			
0	s_1	-4	-2	1	-1	1	0	0	-4	$x_1:\ 1/(1) = 1$	
0	s_2	-1	-1	-1	-1	0	1	0	-5	$x_2:\ 6/(1) = 6$	$\longrightarrow$
0	s_3	2	2	1	1	0	0	1	8	$x_3:\ 2/(1) = 2, \quad x_4:\ 3/(1) = 3$	
	z_j	0	0	0	0	0	0	0	0		
	$z_j - c_j$	1	6	2	3	0	0	0	0		

The most negative RHS is -5: s_2 leaves and x_1 enters on the pivot -1. Divide row 2 by -1, then eliminate column 1 from rows 1 and 3. The new bound is $z_0 = 0 + 1(-5) = -5$.

c_B	Basic	x_1	x_2	x_3	x_4	s_1	s_2	s_3	RHS	Dual ratios ($r = 3$)	Out
	c_j	-1	-6	-2	-3	0	0	0			
0	s_1	0	2	5	3	1	-4	0	16	$x_3:\ 1/(1) = 1$	
-1	x_1	1	1	1	1	0	-1	0	5	$x_4:\ 2/(1) = 2$	
0	s_3	0	0	-1	-1	0	2	1	-2		$\longrightarrow$
	z_j	-1	-1	-1	-1	0	1	0	-5		
	$z_j - c_j$	0	5	1	2	0	1	0	-5		

Row 3 still has RHS -2, so s_3 leaves and x_3 enters on the pivot -1. The next bound is $-5 + 1(-2) = -7$.

c_B	Basic	x_1	x_2	x_3	x_4	s_1	s_2	s_3	RHS
	c_j	-1	-6	-2	-3	0	0	0	
0	s_1	0	2	0	-2	1	6	5	6
-1	x_1	1	1	0	0	0	1	1	3
-2	x_3	0	0	1	1	0	-2	-1	2
	z_j	-1	-1	-2	-2	0	3	1	-7
	$z_j - c_j$	0	5	0	1	0	3	1	-7

Now both invariants hold: every RHS and indicator is nonnegative. Reading nonbasic variables as zero gives

$$x^* = (3, 0, 2, 0), \quad s^* = (6, 0, 0), \qquad \min c^T x = 7.$$

The final tableau also gives the exact identity

$$c^T x = 7 + 5x_2 + x_4 + 3s_2 + s_3 \geq 7. \tag{3}$$

Equality forces $x_2 = x_4 = s_2 = s_3 = 0$. The last two tight primal rows then give $x_1 + x_3 = 5$ and $2x_1 + x_3 = 8$, so this primal optimizer is unique.

3.1 Read and verify the dual certificate

The original minimum has dual

$$\begin{aligned} \max \quad & 4y_1 + 5y_2 - 8y_3, \\ 4y_1 + y_2 - 2y_3 &\leq 1, \\ 2y_1 + y_2 - 2y_3 &\leq 6, \\ -y_1 + y_2 - y_3 &\leq 2, \\ y_1 + y_2 - y_3 &\leq 3, \qquad y \geq 0. \end{aligned}$$

For the transformed equations $[-A, I](x, s) = -b$, the basis multiplier satisfies $-A^T y \geq -c$ and $y \geq 0$, which are precisely these constraints. The last tableau's original surplus indicators give

$$y^* = (0, 3, 1), \quad c - A^T y^* = (0, 5, 0, 1), \quad b^T y^* = 15 - 8 = 7.$$

We have primal and dual feasibility, equal values, and $y^* \odot s^* = 0$, $x^* \odot (c - A^T y^*) = 0$. To see dual uniqueness, $s_1 = 6 > 0$ forces $y_1 = 0$ in every dual optimum. The positive x_1 and x_3 force $y_2 - 2y_3 = 1$ and $y_2 - y_3 = 2$, whose unique solution is $(y_2, y_3) = (3, 1)$.

4 Task 5(a): Phase I proves dual infeasibility

The nonpositive domain of x_1 reverses its dual inequality. The dual is

$$\begin{aligned} \max \quad & 2y_1 + y_2 - 3y_3 + y_4, \\ -y_1 - y_2 - y_3 - y_4 &\geq 3, \\ 2y_1 + y_2 + 2y_3 + 2y_4 &\leq 10, \\ y_1 + y_2 - y_3 + y_4 &\leq 2, \qquad y \geq 0. \end{aligned}$$

Its first condition is impossible because its left side is nonpositive. We can also express this contradiction through the requested two-phase procedure. Introduce surplus t_1 and artificial a_1 in row 1, and slacks s_2, s_3:

$$-y_1 - y_2 - y_3 - y_4 - t_1 + a_1 = 3.$$

Use $B = (a_1, s_2, s_3)$ and maximize $-a_1$. The other two equations have right sides 10 and 2.

c_B	Basic	y_1	y_2	y_3	y_4	t_1	s_2	s_3	a_1	RHS
	c_j	0	0	0	0	0	0	0	-1	
-1	a_1	-1	-1	-1	-1	-1	0	0	1	3
0	s_2	2	1	2	2	0	1	0	0	10
0	s_3	1	1	-1	1	0	0	1	0	2
	z_j	1	1	1	1	1	0	0	-1	-3
	$z_j - c_j$	1	1	1	1	1	0	0	0	-3

Every indicator is already nonnegative. Therefore Phase I has optimum -3, or minimum artificial sum $w = 3$. More directly,

$$a_1 = 3 + y_1 + y_2 + y_3 + y_4 + t_1 \geq 3.$$

A feasible original dual would need $a_1 = 0$, which is impossible. Stop before Phase II: there is no dual optimum and hence no optimal pair to recover.

4.1 A primal ray explains the same outcome

The ray

$$x(t) = (-2 - t, 0, 0), \qquad t \geq 0$$

respects all domains and has primal row values $(2+t, 2+t, 2+t, 2+t)$, which dominate $(2, 1, -3, 1)$. Its objective is $-6 - 3t \to -\infty$. Thus the primal is feasible and unbounded below. This is consistent with the dual contradiction: a feasible dual would supply a finite lower bound and would forbid such a ray. Neither problem has an optimal solution; "alternative optimal dual solutions" therefore do not exist.

5 Task 5(b): solve the dual and recover the primal

Because x_2 is free, its column gives an equality:

$$\begin{aligned} \max \quad & -2y_1 + y_2 - 5y_3 + y_4, \\ 2y_2 - 2y_3 + 2y_4 &\leq 3, \\ -2y_1 - y_2 + y_3 + 2y_4 &= 10, \\ -2y_1 + y_3 + y_4 &\leq 1, \qquad y \geq 0. \end{aligned}$$

Let s_1, s_3 be the slacks of the two dual inequalities, and add a_2 to its equality. Column order is $(y_1, y_2, y_3, y_4, s_1, s_3, a_2)$. Notice that these slacks index dual conditions, not primal resource rows.

5.1 Phase I: three pivots remove the artificial variable

Maximize $-a_2$, starting at $B = (s_1, a_2, s_3)$.

c_B	Basic	y_1	y_2	y_3	y_4	s_1	s_3	a_2	RHS	Ratio ($y_4 \uparrow$)	Out
	c_j	0	0	0	0	0	0	-1			
0	s_1	0	2	-2	2	1	0	0	3	3/2	
-1	a_2	-2	-1	1	2	0	0	1	10	$10/2 = 5$	
0	s_3	-2	0	1	1	0	1	0	1	$1/1 = 1$	$\longrightarrow$
	z_j	2	1	-1	-2	0	0	-1	-10		
	$z_j - c_j$	2	1	-1	-2	0	0	0	-10		

The most negative indicator is $q_4 = -2$. Enter y_4 and remove s_3 after step 1, raising the Phase I value from -10 to -8.

c_B	Basic	y_1	y_2	y_3	y_4	s_1	s_3	a_2	RHS	Ratio ($y_1 \uparrow$)	Out
	c_j	0	0	0	0	0	0	-1			
0	s_1	4	2	-4	0	1	-2	0	1	$1/4$	$\longrightarrow$
-1	a_2	2	-1	-1	0	0	-2	1	8	$8/2 = 4$	
0	y_4	-2	0	1	1	0	1	0	1	–	
	z_j	-2	1	1	0	0	2	-1	-8		
	$z_j - c_j$	-2	1	1	0	0	2	0	-8		

Now y_1 enters because $q_1 = -2$; s_1 leaves after step $1/4$.

c_B	Basic	y_1	y_2	y_3	y_4	s_1	s_3	a_2	RHS	Ratio ($y_3 \uparrow$)	Out
	c_j	0	0	0	0	0	0	-1			
0	y_1	1	$\frac{1}{2}$	-1	0	$\frac{1}{4}$	$-\frac{1}{2}$	0	$\frac{1}{4}$	–	
-1	a_2	0	-2	1	0	$-\frac{1}{2}$	-1	1	$\frac{15}{2}$	$(15/2)/1 = 15/2$	$\longrightarrow$
0	y_4	0	1	-1	1	$\frac{1}{2}$	0	0	$\frac{3}{2}$	–	
	z_j	0	2	-1	0	$\frac{1}{2}$	1	-1	$-\frac{15}{2}$		
	$z_j - c_j$	0	2	-1	0	$\frac{1}{2}$	1	0	$-\frac{15}{2}$		

Finally y_3 enters, with indicator -1, and the artificial a_2 leaves after step $15/2$.

c_B	Basic	y_1	y_2	y_3	y_4	s_1	s_3	a_2	RHS
	c_j	0	0	0	0	0	0	-1	
0	y_1	1	$-\frac{3}{2}$	0	0	$-\frac{1}{4}$	$-\frac{3}{2}$	1	$\frac{31}{4}$
0	y_3	0	-2	1	0	$-\frac{1}{2}$	-1	1	$\frac{15}{2}$
0	y_4	0	-1	0	1	0	-1	1	9
	z_j	0	0	0	0	0	0	0	0
	$z_j - c_j$	0	0	0	0	0	0	1	0

The Phase I value is zero, with a_2 nonbasic and zero. Delete its column; retain the nonsingular basis (y_1, y_3, y_4).

5.2 Phase II and absence of alternative optima

Recompute the objective from $(-2, 1, -5, 1, 0, 0)$; do not reuse the Phase I row.

c_B	Basic	y_1	y_2	y_3	y_4	s_1	s_3	RHS
	c_j	-2	1	-5	1	0	0	
-2	y_1	1	$-\frac{3}{2}$	0	0	$-\frac{1}{4}$	$-\frac{3}{2}$	$\frac{31}{4}$
-5	y_3	0	-2	1	0	$-\frac{1}{2}$	-1	$\frac{15}{2}$
1	y_4	0	-1	0	1	0	-1	9
	z_j	-2	12	-5	1	3	7	-44
	$z_j - c_j$	0	11	0	0	3	7	-44

All indicators are nonnegative, so no Phase II pivot is needed:

$$y^* = (31/4, 0, 15/2, 9), \qquad b^T y^* = -44.$$

For every feasible dual, the tableau implies

$$b^T y = -44 - 11y_2 - 3s_1 - 7s_3 \le -44. \tag{4}$$

A distinct optimizer would have to make an actual feasible move keeping the objective unchanged. Equality forces $y_2 = s_1 = s_3 = 0$; the three basic equations then fix $(y_1, y_3, y_4) = (31/4, 15/2, 9)$. Thus there is no such move and the dual optimum is unique. The zero indicators belong only to basic variables; they do not alone signal alternative optima.

5.3 Recover the primal without a primal simplex solve

Since $y_1, y_3, y_4 > 0$, complementary slackness makes primal rows 1, 3, 4 tight:

$$-2x_2 - 2x_3 = -2, \qquad -2x_1 + x_2 + x_3 = -5, \qquad 2x_1 + 2x_2 + x_3 = 1.$$

The first gives $x_2 + x_3 = 1$. Substitution into the second gives $x_1 = 3$; the third becomes $2x_2 + x_3 = -5$. Therefore

$$x^* = (3, -6, 7), \quad Ax^* - b = (0, 11, 0, 0), \quad c^T x^* = 9 - 60 + 7 = -44.$$

The free variable $x_2 = -6$ is admissible. Both constrained primal variables are nonnegative; the unused row 2 has surplus 11. All dual column slacks are zero. Feasibility and equal objectives certify both optima. The three equations above are independent, so the primal optimum is unique as well.

6 Task 5(c): equality domains reveal infeasibility

Both x_1 and x_2 are free, giving two dual equalities:

$$\begin{aligned} \max \quad & -y_1 + 5y_2 + y_3 - y_4, \\ -y_1 - y_2 - y_3 &= -1, \\ -y_1 + y_3 - y_4 &= 5, \\ 2y_1 + 3y_2 + y_3 - 2y_4 \le 8, \qquad & y \ge 0. \end{aligned}$$

The first equality implies $y_1 + y_2 + y_3 = 1$, hence $-y_1 + y_3 - y_4 \le y_3 \le 1$. It cannot equal 5.

6.1 One Phase I pivot makes the contradiction quantitative

Normalize the first equality by -1, append artificials a_1, a_2 to the first two rows, and use slack s_3 in the third. Maximize $-a_1 - a_2$ with $B = (a_1, a_2, s_3)$.

c_B	Basic	y_1	y_2	y_3	y_4	s_3	a_1	a_2	RHS	Ratio ($y_3 \uparrow$)	Out
	c_j	0	0	0	0	0	-1	-1			
-1	a_1	1	1	1	0	0	1	0	1	$1/1 = 1$	$\longrightarrow$
-1	a_2	-1	0	1	-1	0	0	1	5	$5/1 = 5$	
0	s_3	2	3	1	-2	1	0	0	8	$8/1 = 8$	
	z_j	0	-1	-2	1	0	-1	-1	-6		
	$z_j - c_j$	0	-1	-2	1	0	0	0	-6		

Enter y_3, whose indicator is -2, and remove a_1 after step 1.

c_B	Basic	y_1	y_2	y_3	y_4	s_3	a_1	a_2	RHS
	c_j	0	0	0	0	0	-1	-1	
0	y_3	1	1	1	0	0	1	0	1
-1	a_2	-2	-1	0	-1	0	-1	1	4
0	s_3	1	2	0	-2	1	-1	0	7
	z_j	2	1	0	1	0	1	-1	-4
	$z_j - c_j$	2	1	0	1	0	2	0	-4

All indicators are nonnegative: Phase I stops at $z_{\mathrm{I}} = -4$, hence $w = a_1 + a_2 = 4$. Indeed its second equation reads

$$-2y_1 - y_2 - y_4 - a_1 + a_2 = 4,$$

so

$$a_1 + a_2 = 4 + 2y_1 + y_2 + y_4 + 2a_1 \geq 4.$$

This supplies a hand certificate of the Phase I optimum and the original dual's infeasibility. There is no valid Phase II or recovered optimal primal.

The primal nevertheless has the feasible ray

$$x(t) = (-5 - t, -t, 0), \qquad t \geq 0.$$

Its row values are $(5 + 2t, 5 + t, 5, t)$, dominating $(-1, 5, 1, -1)$, and its objective is $5 - 4t \to -\infty$. Therefore the primal is unbounded below; there are no dual optimal alternatives to discuss.

7 Task 5(d): a feasible dual, two Phase II pivots

Here only x_2 is free. The dual is

$$\begin{aligned} \max \quad & y_1 + 5y_2 + y_3 - 2y_4, \\ 3y_1 - y_2 - y_3 + 3y_4 &\leq 1, \\ y_1 - y_3 - y_4 &= -5, \\ 2y_1 + y_2 + y_3 - 2y_4 &\leq 8, \qquad y \geq 0. \end{aligned}$$

Normalize the equality to $-y_1 + y_3 + y_4 = 5$ before adding artificial a_2; use slacks s_1, s_3. The objective remains the original dual maximum.

7.1 Phase I

Start at $B = (s_1, a_2, s_3)$ and maximize $-a_2$.

c_B	Basic	y_1	y_2	y_3	y_4	s_1	s_3	a_2	RHS	Ratio ($y_3 \uparrow$)	Out
	c_j	0	0	0	0	0	0	-1			
0	s_1	3	-1	-1	3	1	0	0	1	–	
-1	a_2	-1	0	1	1	0	0	1	5	$5/1 = 5$	$\longrightarrow$
0	s_3	2	1	1	-2	0	1	0	8	$8/1 = 8$	
	z_j	1	0	-1	-1	0	0	-1	-5		
	$z_j - c_j$	1	0	-1	-1	0	0	0	-5		

Indicators $q_3 = q_4 = -1$ tie; take y_3 by its smaller index. Remove a_2 after step 5.

c_B	Basic	y_1	y_2	y_3	y_4	s_1	s_3	a_2	RHS
	c_j	0	0	0	0	0	0	-1	
0	s_1	2	-1	0	4	1	0	1	6
0	y_3	-1	0	1	1	0	0	1	5
0	s_3	3	1	0	-3	0	1	-1	3
	z_j	0	0	0	0	0	0	0	0
	$z_j - c_j$	0	0	0	0	0	0	1	0

The Phase I value is zero. Delete the nonbasic artificial. The restored objective coefficients are $(1, 5, 1, -2, 0, 0)$.

7.2 Phase II

c_B	Basic	y_1	y_2	y_3	y_4	s_1	s_3	RHS	Ratio ($y_2 \uparrow$)	Out
	c_j	1	5	1	-2	0	0			
0	s_1	2	-1	0	4	1	0	6	–	
1	y_3	-1	0	1	1	0	0	5	–	
0	s_3	3	1	0	-3	0	1	3	$3/1 = 3$	$\longrightarrow$
	z_j	-1	0	1	1	0	0	5		
	$z_j - c_j$	-2	-5	0	3	0	0	5		

Enter y_2, with indicator -5, and remove s_3 after step 3. The objective rises from 5 to 20.

c_B	Basic	y_1	y_2	y_3	y_4	s_1	s_3	RHS	Ratio ($y_4 \uparrow$)	Out
	c_j	1	5	1	-2	0	0			
0	s_1	5	0	0	1	1	1	9	$9/1 = 9$	
1	y_3	-1	0	1	1	0	0	5	$5/1 = 5$	$\longrightarrow$
5	y_2	3	1	0	-3	0	1	3	–	
	z_j	14	5	1	-14	0	5	20		
	$z_j - c_j$	13	0	0	-12	0	5	20		

Now enter y_4, with indicator -12, and remove y_3 after step 5. The objective rises by $12 \cdot 5 = 60$.

c_B	Basic	y_1	y_2	y_3	y_4	s_1	s_3	RHS
	c_j	1	5	1	-2	0	0	
0	s_1	6	0	-1	0	1	1	4
-2	y_4	-1	0	1	1	0	0	5
5	y_2	0	1	3	0	0	1	18
	z_j	2	5	13	-2	0	5	80
	$z_j - c_j$	1	0	12	0	0	5	80

This yields

$$y^* = (0, 18, 0, 5), \qquad b^T y^* = 90 - 10 = 80.$$

The final identity

$$b^T y = 80 - y_1 - 12y_3 - 5s_3 \le 80 \tag{5}$$

forces $y_1 = y_3 = s_3 = 0$ at every optimum. The remaining equations give $y_4 = 5$, $y_2 = 18$, $s_1 = 4$. No nonzero feasible displacement can keep the objective at 80, so the dual optimum is unique.

7.3 Primal recovery and the role of the free variable

The dual column slacks are

$$c - A^T y^* = (4, 0, 0).$$

Because $x_1 \geq 0$, complementarity gives $x_1 = 0$. Positive y_2, y_4 make primal rows 2 and 4 tight:

$$-x_1 + x_3 = 5, \qquad 3x_1 - x_2 - 2x_3 = -2.$$

Hence $x_3 = 5$, $x_2 = -8$. The recovered point is

$$x^* = (0, -8, 5), \qquad Ax^* - b = (1, 0, 12, 0), \qquad c^T x^* = 40 + 40 = 80.$$

It obeys the domains, both inactive inequalities have nonnegative surplus, and the primal and dual values agree. These equations force the same point for every primal optimum, proving its uniqueness without a primal simplex run.

8 Interpretation and common mistakes

- **Objective signs:** task 4 negates the primal objective for its dual-simplex tableau; report 7 as the original minimum, not -7. Tasks 5 maximize the dual directly, including the valid negative maximum -44 in 5(b).
- **Free and nonpositive domains:** a free primal coordinate creates an equality in the dual. A nonpositive coordinate reverses the column inequality. Bounds on the right side do not determine these signs.
- **Two-phase transition:** artificials are feasibility devices. Delete their columns at a zero Phase I optimum and recompute the original objective. A negative Phase I maximum forbids Phase II.
- **Alternative optima:** a zero indicator is useful only with an actual feasible move. In 5(b) and 5(d), every nonbasic indicator is strictly positive, and the basic equations leave no optimal movement.
- **Primal recovery:** solve the equations forced by a dual optimizer, then verify all remaining inequalities and domains. Complementarity alone does not establish feasibility.
- **Unboundedness:** give a feasible starting point and an improving ray. An infeasible dual alone would not prove primal unboundedness if primal feasibility had not also been established.

9 MATLAB comparison

The companion `sheet_01_dual_simplex_and_recovery.m` uses the same printed matrices, augmented equations, basis orders, and pivot rules as the hand solutions. From the repository root, run

```
run('matlab/exercises/sheet_01_dual_simplex_and_recovery.m')
```

It uses base MATLAB linear solves, with absolute tolerance 10^{-10} for residuals, tableau comparisons, sign tests, and objective values. This tolerance is appropriate to these small inputs; it is not a conditioning guarantee for arbitrary models. The script checks the specified paths, including objective-row recomputation, independently against new basis solves. It checks primal/dual feasibility, complementarity, positive nonbasic indicators for uniqueness, and feasible rays for the unbounded cases. It does not implement a general LP solver.

Task	Primal outcome	Dual outcome	Phase I artificial sum
4	Unique minimum 7	Unique maximum 7	Not used
5(a)	Unbounded below	Infeasible	3
5(b)	Unique minimum -44	Unique maximum -44	0
5(c)	Unbounded below	Infeasible	4
5(d)	Unique minimum 80	Unique maximum 80	0

The table records exact expected hand results, not invented solver output. Mathematical, static-analysis, and MATLAB execution evidence is identified separately in the final remarks and repository verification record.

10 Sources and editorial remarks

Task source: `../execrices-2026-27/ProgMatc1.pdf`, PDF page 2, tasks 4 and 5(a)–5(d). The Polish instructions and mathematical data are preserved in Section 1; layout is retypeset. The supplied current-course context is 2026–27. The PDF creation metadata is 2025, which does not establish the delivery date. No lecture date is claimed.

The hand derivations, complete tableaus, recovery equations, rays, and uniqueness arguments are explanatory additions. The generic instruction in task 5 requests optimal solutions. Sections 4 and 6 show that the printed domains and coefficients instead imply unbounded primal problems and infeasible duals. The task statements remain unchanged; their intended variants are unknown. Successful Phase II solves are provided for 5(b) and 5(d).

Verification status: hand tableaus and pivot choices were independently recomputed using exact rational arithmetic. The user supplied successful MATLAB output on 2026-10-11 and confirmed version 26.2.0.3386108 (R2026b). All dual-simplex and recovery assertions passed, including the Phase I failures and primal rays in 5(a) and 5(c). The supplied execution evidence is retained in `docs/matlab-verification.md`.